INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

BLAST CLEANING MACHINE: SELF-CONTAINED TUMBLE TYPE, WITH DUST COLLECTOR

This specification is approved for use by the Armament Research, Development and Engineering Center (Tool and Equipment Engineering Division, Standardization and Specification Engineering Branch), Department of the Army and is available for usage by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a self-contained tumble type blast cleaning machine, including dust collector, in which the abrasive is directed by centrifugal force onto the objects being cleaned.

1.2 Classification. The blast cleaning machines covered by this specification shall be the following sizes, as specified (see 6.2):

Size 1-1 1/2 cubic feet minimum (covering 1-1/2 cu. ft. machines)
Size 2-3 cubic feet minimum (covering 3 to 5 cu.ft. machines)
Size 3-6 cubic feet minimum (covering 6 to 7 cu. ft. machines)
Size 4-12 cubic feet minimum (covering 12 to 14 cu. ft. machines)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament, Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4940

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SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
P-D-680	-	Dry Cleaning and Degreasing Solvent
TT-P-664	-	Primer Coating, Synthetic, Rust-Inhibiting, Lacquer Resisting
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Box, Wood, Nailed and Lock-Corner
PPP-B-636	-	Box, Shipping, Fiberboard
PPP-B-640	-	Box, Fiberboard, Corrugated, Triple-Wall
PPP-B-1055	-	Barrier Material, Waterproof, Flexible
PPP-C-843	-	Cushioning Material, Cellulosic
PPP-D-705	-	Drum, Shipping and Storage: Steel, 16 and 30 Gallon Capacity
PPP-D-729	-	Drum, Shipping and Storage, Steel, 55-Gallon (208 Liters)
PPP-P-40	-	Preservation and Packing of Hand Tools; Tools and Tool Accessories For Power Driven, Metal and Woodworking Machinery
PPP-T-60	-	Tape: Packaging, Waterproof

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MIL-C-104	-	Crate, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	-	Preservation, Methods of
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-P-775	-	Packaging of Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing); and Fittings, Nozzles, and Strainers
MIL-C-3774	-	Crates, Wood; Open 12,000- and 16,000- Pound Capacity
MIL-C-5501	-	Caps and Plugs, Protective, Dust and Moisture Seal
MIL-H-6083	-	Hydraulic Fluid, Petroleum Base, For Preservation and Operation
MIL-S-12514	-	Starters, Motor; Across-the-Line and Reduced Voltage Types, A-C Induction, 3/4 to 100 HP, Manual and Electric
MIL-E-16298	-	Electric Machines Having Rotating Parts and Associated Support Items: Packaging of
MIL-E-17555	-	Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of
MIL-B-26195	-	Boxes, Wood-Cleated, Skidded, Load-Bearing Base
MIL-C-52950	-	Crates, Wood, Open and Covered

STANDARDS

FEDERAL

FED-STD-H28	-	Screw Thread Standards For Federal Services Federal Government
FED-STD-101	-	Test Procedures for Packaging Materials (Method 5011.1)

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MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of US Military Property
MIL-STD-889	-	Dissimilar Metals
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking and Waterproofing with appropriate Test Methods
MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of federal and military specifications are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those cited in the solicitation.

US DEPARTMENT OF LABOR

29 CFR 1910, Chapter XVII, Part 1910	-	General Industry, OSHA Safety and Health Standards

(Unless otherwise indicated, copies of the regulation are available from the Superintendent of Documents, Government Printing Office, Washington DC 20402.)

2.2. Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issue of the document cited in the solicitation.

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

AGMA 390.03	-	Gear Handbook, Volume I, Gear classification, Materials, and Measuring Methods for Unassembled Gears
AGMA 2000-A88	-	Gear Classification and Inspection Handbook

(Application for copies should be addressed to American Gear Manufacturers Association, One Thomas Circle, Washington, DC 20005.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA ICS 1	-	General Standards for Industrial Controls and Systems
NEMA MG 1	-	Motors and Generators

(Applications for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, NW, Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	-	National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents are also available in or through libraries or other informational sevices.

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.)

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.2 Design. The blast cleaning machine shall be new and of the latest standard commercial design meeting the performance requirements of this specification. The blast cleaning machine shall be a self-contained tumble type with a load capacity of not less than the minimum cubic feet shown for the size specified in Table I. The machine shall house an endless belt conveyor upon which the parts to be cleaned are loaded, tumbled while being cleaned, and unloaded. The machine, with the door closed, shall be dust and abrasive tight. The blast action shall be accomplished by a controlled centrifugal force acting upon the abrasive. An abrasive recovery and handling system shall be provided as an integral part of this unit. The dust collector shall be a separate cabinet with its own power source (motor). All parts subject to wear or replacement shall be accessible for adjustment and repair when using standard tools and maintenance equipment.

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with best commercial practice, shall be suitable for the intended purpose in design and fabrication of the end item, and shall meet all the requirements herein. Materials shall be free from defects which would affect the performance, maintainability, reliability, durability, or longevity of the individual components or the end item of equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.1 Reclaimed materials. The manufacturer may use reclaimed materials for the fabrication of new parts. Reclaimed materials shall be reprocessed, remanufactured, or recycled in a manner that restores them to the same chemical composition and physical properties as the raw materials originally selected for use. Use of reclaimed parts as is or rebuilt from scrap or other used equipment shall not be permitted.

3.4 Construction. The blast cleaning machine shall be complete, and shall be capable of operating in accordance with the requirements specified herein. All parts used in fabrication shall be new. There shall be no process used for reclaiming any parts of the blast cleaning machine. All surfaces of all parts shall be clean and free from extraneous materials. External surfaces shall be smooth and free of sharp edges. All screws, pins, bolts, and similar locking parts shall be installed with a means for preventing loss of tightness. All such parts, when subject to removal or adjustment, shall not be swagged, peened, staked, or otherwise permanently deformed.

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3.4.1 Interchangeability. All parts of the end item shall be manufactured to definite standards and tolerances that will provide for the interchangeability of respective parts between end items of the same model. All replacement parts shall be interchangeable with the respective original parts, and fit and function with mating components.

3.4.2 Threads. All machined threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.4.3 Electrical connections. Connections of conductors and terminal parts shall be of the screw, pressure, or solder type. When soldered connections are used, the conductors and terminal parts shall be mechanically secured before soldering. Rosin base fluxes only shall be used in soldering operations. Connections to screw-type terminals shall be mechanically secured with a means to prevent loss of tightness.

3.4.4 Gears. Gears and pinions used in the blast cleaning machine or its components shall be fabricated in either the English (USA) system or the Metric (SI) system of measurement. Gears shall conform to AGMA 390.03 or 2000-A88 as applicable.

3.4.5 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with standard means for lubrication. Oil holes, grease fittings, and filler caps shall be accessible. Lubrication reservoirs, if applicable, shall have means for checking levels.

3.4.6 Safety provisions. Covers, guards, or other safety devices shall be provided with the blast cleaning machine. The devices shall not interfere with the operation of the blast cleaning machine. As a minimum, all moving parts located on the blast cleaning machine from floor level to a height of eight feet shall be enclosed. Protection against electrical shock shall be provided. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All blast cleaning machine parts, components, mechanisms and assemblies furnished whether or not specifically required herein, shall comply with all of the requirements of 29 CFR, Chapter XVII, Part 1910 that are applicable to the blast cleaning machine itself. Special safety devices shall be provided when specified (see 6.2).

3.5 Performance and product characteristics.

3.5.1 Function of blast cleaning machine. The blast cleaning machine shall be capable of performing a predetermined blast cleaning cycle. This performance shall consist of tumbling and cleaning a work load while circulating a metallic abrasive. The work load shall be tumbled on an endless belt conveyor. The circulating rate of the metallic abrasive flow shall be accomplished through the combined facilities of an airless blasting unit, the endless belt conveyor, and an elevator. The minimum abrasive capacity of the cleaning machine shall be sufficient to maintain the minimum rate of flow.

3.5.2 Function of dust collector. The dust collector shall be capable of filtering dust particles from the air inside the cleaning cabinet so the exhaust air shall not contaminate the air in the building. A shaking device shall remove the dust from the tubes when the air flow through the dust collector is shut off.

3.6 Detail of components. The blast cleaning machine shall consist of, but not be limited to, the following component parts.

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3.6.1 Machine cabinet and frame. The machine cabinet and frame shall accommodate an endless belt conveyor, the drive mechanism, an airless blasting unit, the abrasive handling and reclaim system, and motors. The blast cleaning machine shall be mounted at floor level. The machine cabinet shall be fabricated from steel plates or panels not less than Manufacturers Standard Gage No. 14. These plates or panels shall be welded on a steel frame. The section surrounding the blast cleaning chamber shall be dust and abrasive tight. Inspection doors or plates shall be provided for access to all serviceable parts or assemblies. The cabinet and frame shall be capable of supporting a minimum of 3,000 pounds, which includes the cleaner components, work load, and abrasive load, without showing any evidence of damage. An instruction plate shall be located on the front of the machine cabinet. These instructions shall include the proper operating and service procedures for the blast cleaning machine.

3.6.1.1 Cabinet work door. The cabinet work door shall be dust and abrasive tight when closed. The opening and closing of the door may be either manually or power operated. The door shall be provided with a safety inner-lock switch, which stops the blasting unit when the door is open and prevents the conveyor from reversing when the door is closed. The door shall be provided with a lock, which locks automatically when the door is closed and has to be manually or electrically unlocked before the door can be opened.

3.6.2 Airless blasting unit. The airless blasting unit shall be provided with a cycle control plug or valve, which shall stop the blasting unit and abrasive flow at the end of the blast cycle or when any of the other cleaning machine operations are interrupted. The blasting unit shall be driven by a motor of the horsepower (hp) specified in table I, and be fan-cooled with multiple V-belt drive. The blasting unit shall consist of the following principal components.

3.6.2.1 Shaft and bearing assembly. The shaft and bearing assembly shall be of dust proof construction. The shaft and bearing assembly shall consist of anti-friction bearings, shaft, and dust proof oil seals appropriate to the design and mounting.

3.6.2.2 Wheel or runner assembly. The wheel or runner assembly shall consist of the blades or vanes, an impeller, and a control cage mounted on the shaft and bearing assembly. A minimum of eight replaceable blades or vanes shall be secured to the hub or side plates of the wheel or runner assembly.

3.6.2.2.1 Impeller and control cage. The impeller and control cage shall be mounted on the inside diameter of the wheel to distribute abrasive to the blades or vanes. The abrasive shall be fed by gravity into the impeller and, from there, the impeller paddles shall discharge the abrasive through slots in the control cage onto the blades or vanes, which direct the abrasive at high velocity against the tumbling work pieces.

3.6.2.3 Wear plates. The inside of the wheel or runner housing shall be completely lined with replaceable wear plates. The inside ends of the drum or barrel heads shall be lined with replaceable wear plates. All wear plates shall be of a wear-resistant material with at least 50 times the life of mild steel.

3.6.2.4 Abrasive flow and direction control. The rate of flow and the direction of the abrasive for blast cleaning shall be capable of being controlled by the operator. The control device shall be mechanical and shall be accessible for adjustment by the operator.

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3.6.3 Endless belt conveyor. The endless belt conveyor shall be made of an abrasive-resisting metal or of a molded rubber base material, as specified (see 6.2). If a rubber belt conveyor is used, a mechanically connected belt is not acceptable. Holes of 1/4 inch nominal diameter shall be provided in the belt to allow the abrasive to fall into the hopper and the abrasive handling mechanism. Adjustments for tension and alignment of the belt conveyor shall be made from the outside of the cabinet. A totally enclosed, dust tight, safety mechanism shall be provided to stop the blasting unit and the belt conveyor if work is jamming or if the belt conveyor is overloaded.

3.6.3.1 Work load capacity. The maximum work load capacity of the belt conveyor shall be as specified in Table I. The belt conveyor shall also be able to withstand the impact of pieces weighing 25 pounds. Holding and tumbling the work load shall be done without evidence of strain or damage to the blast cleaning machine or any of its components.

3.6.3.2 Conveyor pulleys and guides. The conveyor pulleys and guides shall be mounted on the machine frame, and shall be constructed to prevent the abrasive from clogging or building up between the pulleys and belt conveyor.

3.6.3.3 Conveyor drive. The belt conveyor shall be driven by the applicable motor specified in Table I. The conveyor drive shall run from the motor through a reducer unit to the rear conveyor shaft. The drive shall be protected from direct blast of the abrasive.

3.6.4 Drum or barrel heads. A drum or barrel head shall be located on each side of the belt conveyor to guide its travel and to form the contour of the blasting chamber, where the work load is held and tumbled for cleaning. The heads shall be driven by the belt conveyor. The heads shall be made of heavy steel and shall have provisions for securing wear plates to the inside faces. The drum or barrel head hubs shall be mounted on heavy duty antifriction type bearings in dust-tight housings.

3.6.5 Abrasive handling system. The abrasive handling system shall retrieve the used abrasive from beneath the belt conveyor and return it to the storage hopper, completing the cycle for abrasive cleaning. The abrasive handling system shall consist of an elevator and drive mechanism, a removable wire screen abrasive separator, and a storage hopper.

3.6.5.1 Elevator and drive mechanism. A bucket-type elevator shall carry the abrasive from the hopper, which is located beneath the belt conveyor and screen, to the separator. From here the abrasive shall fall into the abrasive storage hopper. The elevator shall be motor driven by a standard V-belt unit or appropriate reduction drive unit. The elevator tension and alignment shall be adjustable without requiring disassembly of the elevator or any of its components.

3.6.5.2 Removable wire screen. A removable wire screen shall be provided between the apron conveyor and the abrasive hopper for removing all course refuse from used abrasive.

3.6.5.3 Abrasive separator. An abrasive separator shall be located between the storage hopper and where the abrasive is emptied from the elevator buckets. The abrasive separator shall separate the dust, refuse, and abrasive fines from the usable abrasive. The dust and abrasive fines shall be exhausted by the dust collector, and the coarser refuse shall be diverted from entering the abrasive

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TABLE I. Size and requirements.

CHARACTERISTIC	REQUIREMENT			
	Size 1	Size 2	Size 3	Size 4
Load capacity (cubic feet), not less than	1.5	3	6	12
Work load (pounds), not less than	225	400	1,000	1,500
Metallic abrasive flow (pounds per hour)	4,500	15,000	25,000	40,000
Max. work load capacity of belt conveyor (pounds)	225	400	1,000	1,500
Conveyor drive motor	1/3 hp	1/2 hp	3/4 hp	1 hp
Elevator drive motor	1/2 hp	3/4 hp	1 hp	1-1/2 hp
Storage hopper capacity, abrasive flow (pounds per hour)	4,500	15,000	25,000	40,000
Exhaust fan motor	2 hp	3 hp	4 hp	5 hp
Air-cloth-ratio (cu. ft. per min.)	3.5 to 1	4.5 to 1	5.5	6.5
Outlet velocity of air from the exhauster (Feet per minute)	2,400	3,300	4,300	5,300
Cloth dust filters (square feet)	300	350	450	550
Airless blasting unit motor size at least horsepower	5	7.5	15	20

storage hopper and shall be collected in another area in the cabinet. Any mechanical action required by the separator shall be provided by the elevator drive mechanism.

3.6.5.4 Storage hopper. The storage hopper shall be mounted over the airless blasting unit so that the abrasive is gravity fed to the blasting unit. The storage hopper shall have an adjustable orifice for regulating the abrasive rate of flow to the blasting unit. The storage hopper shall have the capacity to maintain a minimum abrasive flow to the airless blasting unit as specified in Table I.

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3.6.6 Dust collector. Unless otherwise specified (see 6.2), a dust collector shall be supplied with each machine. The dust collector shall be the suction type and shall consist of, but not be limited to, the following principal component parts.

3.6.6.1 Dust collector cabinet and frame. The dust collector cabinet shall have a steel frame capable of supporting of at least 2,500 pounds. The cabinet panels shall be fabricated from sheet steel not less than Manufacturers Standard Gage No. 14. The cabinet shall provide access to the dust filters for periodic inspection and replacement. The cabinet area enclosing the dust filters shall be dust tight.

3.6.6.2 Cloth dust filters. The total area of the cloth dust filters shall be not less than the area specified in Table I. The dust filters shall be attached to the shaking mechanism so that no part of the filtering area will touch any part of the cabinet or mechanism.

3.6.6.3 Filter shaking mechanism. The shaking mechanism which requires a drive motor shall be equipped with not less than 1/3 horsepower. The shake mechanism shall be mounted on top or side of the cabinet and shall effectively remove dust during shaking without causing material damage to the filters.

3.6.6.4 Dust hopper. The dust hopper shall be mounted directly beneath the dust filter cabinet with an 18 inch clearance from the floor. Only updraft type filter systems shall require a screen or baffle which shall prevent coarse material from being drawn into the filter before dropping into the hopper. Dust and all foreign refuse shall be removed by manually opening a locked gate at the bottom of the hopper.

3.6.6.5 Exhaust fan and motor. The exhaust fan and motor may be mounted on top of the cabinet. The exhaust fan motor shall be as specified in Table I. The exhaust fan and motor shall be capable of handling at least 1,050 cubic feet of air per minute (cfm) at 6 inches or water column static pressure. The air-cloth-ratio shall not exceed that specified in Table I at the maximum cfm. The outlet velocity of the air from the exhauster shall be as specified in Table I. The motor and exhaust fan shall be provided with a multiple V-belt drive when belts are required to transmit power. The motor and exhaust fan may use direct drive without belts. Updraft type filter systems shall be provided with an interrupt circuit which shall prevent simultaneous operation of the shake mechanism and exhauster. Downdraft filter systems shall be provided with means for breaking negative pressure at intervals during continuous operation of the exhauster.

3.6.6.6 Operating controls. The operating control panel shall include all the electrical controls with their function properly marked. The wiring circuit and components within the control panel shall conform to NEMA ICS 1 and NFPA 70 as applicable.

3.6.6.7 Instruction plate. An instruction plate shall be located on the front of the dust collecting cabinet. The instructions shall include the proper operating and service procedures for the dust collector.

3.6.7 Electrical components. Unless otherwise specified (see 6.2), the blast cleaning machine shall be completely wired when furnished. Electrical components, including starters, relays, switches, wiring, and their enclosures, shall conform

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to, and shall be located in accordance with, NEMA ICS 1 and NFPA 70 as applicable. There shall be no component part used in any application where the requirements herein imposed on the part exceed the manufacturer's rating for the part.

3.6.7.1 Motors. Unless otherwise specified (see 6.2), all motors shall be the induction, squirrel-cage type, totally enclosed, dust tight, and operate from 230/460 volts, 60 hertz (Hz), alternating current (ac), 3-phase, conforming to the requirements of NEMA MG 1 with class B insulation and continuous duty rating. When starters are required, they shall conform to type I, class 2 requirements of MIL-S-12514. Motors shall be initially wired for 230 volts.

3.6.7.2 Push button circuits. Push button stations shall be provided for each starter, and shall be located on the cabinet within easy reach of the operator. All push button circuits shall be 115 volt. Each motor shall be provided with an individual dustproof push button switch.

3.7 Spare parts. When specified (see 6.2), the following spare parts shall be furnished with each machine in the quantity specified:

Item	Quantity
a. Vanes or blades for blasting unit	One set
b. Wear plates for blasting unit	One set
c. Wear plates for drum or barrel heads	One set

3.8 Painting and protective finish. Painting and protective finish shall be in accordance with the best practice of commercial industry. The protective finish shall be a paint coating of at least 2.0 mils (0.0002 inch) thickness. All surfaces to be painted shall, immediately prior to painting, be cleaned, be free of all foreign matter, and be dried. Paint shall adhere to the surface of the end item, when fiberglass tape is removed from two intersecting knife cuts, at least two inches long, through to the base metal. The surface hardness of the paint shall be between the industrial standards of H to HB pencil lead. Where no protective finish is utilized (such as bedways, Machine guide surfaces, etc.) the natural finish of the material or the finish obtained from heat treatment is permissible provided the surfaces are free of scale and have a protective coating to inhibit corrosion.

3.9 Product identification. The blast cleaning machine shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2), shall include the National Stock Number.

3.10 Workmanship. Standards of workmanship shall assure that the blast cleaning machine shall have the stability and efficient operating characteristics found in standard commercial items and as specified in Section 3 herein.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production units being offered for acceptance under this specification. Unless otherwise specified (see 6.2), each blast cleaning machine shall be subjected to the examinations and tests in 4.5, 4.6.1 through 4.6.4, and 4.7. If a first article test (FAT) is not specified, one blast cleaning machine and dust collector unit from the first lot shall be subjected to the packaging validation tests in 4.6.5. The test(s) in 4.6.5 do not need to be repeated for quality conformance inspection if the test(s) in 4.6.5 have been conducted in a FAT and the same equipment is used for the packaging validation tests. Failure of any device to pass these inspections or tests shall be cause for rejection.

4.4.1 Sampling. Sampling for quality conformance inspection shall be performed as listed with all samples selected randomly.

Inspection or test	Lot or batch size	Sample size
Product examination (4.5)	1-90	13
Tests (4.6.1 through 4.6.4)	1-90	13
Packaging inspection (4.7)	1-90	3
Packaging validation test (4.6.5)	1/	-

The lots shall not exceed the maximum sizes indicated above. If lot size is less than or equal to sample size, 100 percent inspection is required. Each lot shall be accepted with no defects and rejected if one or more defects are found.

1/ The packaging validation tests shall be conducted on one of the packaged items, or one set of items if packaged separately, selected from the first lot submitted for quality conformance inspection.

4.5 Product examination. The blast cleaning machine shall be visually, manually, and dimensionally examined to determine conformance with the requirements of 3.2 through 3.4.6, 3.6 through 3.6.2.4, 3.6.3.2 through 3.7, and 3.9 through 3.10 Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning,

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and freedom from the identified defects. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. Dimensional examination shall verify size and weight conditions when specified. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the blast cleaning machine and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item, shall be cause for rejection (see 3.4.1).

4.6 Tests.

4.6.1 Loading to capacities. Before testing, the blast cleaning machine shall be loaded for operation. The belt conveyor shall be loaded with the applicable maximum work load specified in Table I. The abrasive handling system shall be loaded with metallic abrasive to the applicable minimum capacity that shall maintain the rate of flow shown in table I. Failure of the machine to maintain the minimum abrasive flow rate shown in table I shall be cause for rejection.

4.6.1.1 Conveyor load capacity. The belt conveyor shall be operated through a cycle that includes loading the conveyor with 25 pound $\pm$ 2 pound pieces to the maximum work load capacity, agitating the work load, opening and closing the work door, and reversing the belt to unload the conveyor, to determine compliance with 3.6.3.1.

4.6.1.2 Abrasive rate of flow. While the blast cleaning machine is in operation, a measurement of the abrasive rate of flow between the storage hopper and the airless blasting unit shall be taken to determine compliance with 3.6.3.

4.6.2 Functional test of blast cleaning machine. The blast cleaning machine shall be loaded in accordance with 4.6.1. The machine shall then be operated through a complete blast cleaning cycle including working all the principal components, controls, valves, switches, and other working parts necessary in order to determine compliance with 3.5.1.

4.6.3 Functional test of dust collector. The dust collector shall be connected to the blast cleaning machine with a minimum distance of 20 feet between the two units. The dust collector functional test shall be run simultaneously with that of the blast cleaning machine. The dust collector shall exhaust from the cleaning machine and discharge the filtered air at the applicable velocity specified in table I; without contaminating the air in the building; and meeting any noxious vapors, health, and safety requirements specified in the contract (see 6.5). The shaking device shall be actuated to remove the dust from the tubes after the exhauster has been shut off (see 3.5.2).

4.6.4 Paint test. A test sample, made of the same metal as the production item, shall be tested for paint adhesion, hardness, and thickness. The sample shall receive the same surface treatment, the same thickness of paint, and be painted at the same time with the same paint as the production item. The tests are described in a, b, and c following. The criteria for rejection are given in d.

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a. Two intersecting straight lines, approximately two inches long, shall be cut through the paint to the base metal of the test sample. A strip of 1/2 inch fiberglass tape shall be pressed over the intersection and pulled from the surface.

b. The sharpened leads of an H and HB wooden lead pencil shall be pressed vertically onto the painted surface until the leads crumble or break. The surface shall be cleaned and examined.

c. The thickness of the unpainted test sample shall be measured and recorded to an accuracy of 0.0002 inch. The measured value shall be used in the calculation of final paint thickness.

d. A paint thickness of less than 2.0 mils (0.0020 inch), or an indentation caused by the HB lead, or no indentation left by the H lead, or evidence of any paint loss due to removal of the fiberglass tape, shall be cause for rejection (see 3.8).

4.6.5 Packaging validation tests. After the quality assurance inspection and tests have been successfully completed, one blast cleaning machine unit, packaged in accordance with the requirements of Section 5, shall be selected at random and subjected to the following rough handling tests specified in FED-STD-101. If a FAT is required, the packaging validation test shall be conducted on the epuipment used for FAT. If a FAT is not required by, then the equipment from the first production lot used for the tests in 4.6.1 through 4.6.4 shall be used for the packaging validation test. The apparatus used shall be as described in FED-STD-101. A dummy load shall not be used. All tests are to be conducted under ambient conditions. Environmental temperature - humidity tests are not required. After the test has been completed, the container shall be inspected. Defects in accordance with those described below shall be cause for rejection. The unit shall be subjected to the quality assurance inspections in 4.5 to reaffirm that the blast cleaning machine meets the requirements of Section 3. The tests in 4.6.1 through 4.6.4 do not need to be repeated after the packaging validation test. After testing, any damage to the packaging shall be replaced prior to the acceptance of the lot.

4.6.5.1 Mechanical handling test. The lifting and transporting by forklift truck test,shall be conducted using method 5011.1 of FED-STD-101. Criteria for passing or failing of the test is as follows:

"When the container or package is subjected to the mechanical handling test, the contents (except a dummy load) shall show no functional or physical damage, and the container and packing shall show no functional damage. Damage of the exterior shipping container which is the result of improper interior, packaging, blocking, or bracing shall be cause for rejection. Structural failure of the exterior shipping container which would result in spilling of the contents or failure of the container in subsequent handling is cause for rejection. There shall be no evidence of a substantial amount of shifting of the contents within the exterior shipping container that would create conditions likely to cause damage during shipment, storage, and reshipment of the container. Minor container damage such as chipping of wood members, negligible dents, paint chipping, is not cause for rejection."

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

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5 PACKAGING

5.1 First article. When specified (see 6.2), the supplier shall furnish one complete pack for first article inspection. Whenever first article inspection is specified or performed, unit pack examination and tests for packaging may be performed also (see 4 and 5.10). The first article may be either a preproduction pack or an initial production pack which conforms to the requirements of this specification. In either case, the approved first article and the production packs shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment and packaging in accordance with the requirements of this specification.

5.2 Disassembly. When practical, for the protection of components, attachments and accessories against damage, pilferage or to reduce cubage, items may be disassembled only to a necessary basic degree. Disassembled parts shall be preserved, anchored, braced, blocked, and cushioned to prevent damage. Disassembled parts shall be clearly and legibly marked as to identity and proper location on the assembled item. All fasteners removed during disassembly shall be secured in one of the mating parts. A part shall not be removed from an assembly unless it can be reassembled readily in the field without special skills or tools.

5.3 Disassembled parts marking. Disassembled parts shall be marked, when necessary with instructions, to facilitate reassembly. Removed parts and mating parts on the item shall be marked by stenciling, on the part, or by use of tags. Tags shall be waterproof and ink used (on parts or tags) shall be waterproof. Tags shall be firmly secured to each individual part.

5.4 Materials. Materials used in preservation-packaging and packing of machinery shall be as specified herein. Materials shall be new and free from all defects and imperfections which may affect the serviceability of the completed package or pack.

5.5 Processing conditions. Preservation-packaging shall be accomplished within buildings which have facilities to limit the accumulation of dust and moisture on the equipment being processed. Machinery brought into processing areas from outside shall be allowed sufficient time to reach temperature before processing. Cleaning, drying, and application of preservatives shall progress in an interrupted series of operations. When periods of interruption are necessary, temporary protection shall be provided to partially processed items, as required, to insure against contamination.

5.6 General requirements. Components parts of machinery, accessories, and maintenance tools shall be cleaned, dried, preserved, packaged, and packed as specified herein. Parts and assemblies which are not covered by this specification shall be preserved and packaged utilizing the materials and methods specified for similar parts and assemblies described herein. Cleaning and drying procedures and preservatives and their application, specified herein, shall conform to the criteria and guidelines of MIL-P-116

5.7 Preservation-packaging. Preservation shall be Level A or C, as specified in the contract (see 6.2).

5.7.1 Level A.

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5.7.1.1 Preservation-packaging. Unless otherwise specified, the basic machine shall be preserved and packaged in accordance with method I criterion of MIL-P-116. Electronic and electrical equipment, hand tools and accessories shall be preserved and packaged as specified herein. When it has been determined that a preservative is required to protect a specific surface or item, but has not been specified herein, the preservative utilized shall be selected from MIL-P-116.

5.7.1.2 Cleaning and drying. Thorough cleaning and drying of all bare ferrous metal surfaces is essential before the application of a preservative or other protective finish such as paint, plating, etc.. Care must be taken to limit the handling of cleaned parts prior to the assembly of new machinery. Where practicable, processors and assemblers shall wear rubber or plastic gloves while handling cleaned parts. Unless otherwise specified herein, machines which have been functionally tested prior to shipment shall have all operational systems thoroughly drained before the application of a preservative. It may be necessary to connect the machine to a power source and operate at low speeds in order to drain all cavities. Extreme care shall be taken to immediately disconnect power when machine is being drained so as not not to damage pumps, motor or other components.

5.7.1.3 Internal surfaces. All ferrous metal surfaces of parts which will be utilized internally in the assembly of new machinery shall be cleaned in accordance with process C-5 and dried in accordance with the applicable procedure of MIL-P-116. In the interim, to preclude contamination from forming on cleaned surfaces prior to the assembly operation, parts shall be coated with a light weight preservative oil conforming to P-9. To prevent dust from accumulating on cleaned parts, cover with a sheet of greaseproof material, such as MIL-B-121. In the event the supplier deems it necessary to clean a particular mechanism after draining the operational lubricant, flush with a solution of one part lubricating oil P-10, type I, grade 30 and nine parts cleaning solvent conforming to P-D-680. Otherwise, the mechanism need not be cleaned. All ferrous metal internal surfaces shall be coated with the lubricant or a preservative. Internal mechanisms shall be cycled to assure all areas are protected with a coating of the lubricant before either sealing or draining. Any unprotected areas will have a preservative coating applied.

5.7.1.4 External surfaces. All accessories and attachments such as chucks, faceplates, steady rests, toolholders, and abrasive wheels shall be removed from the assembly prior to the external cleaning operation. All external critical ferrous metal surfaces (fit or function) of the machine shall be cleaned in accordance with process C-5 and dried in accordance with the applicable procedures of MIL-P-116. Noncritical ferrous metal surfaces shall be cleaned in accordance with process C-3 and dried as specified above. Nonmetallic composites and painted metal surfaces shall be cleaned by wiping with a clean, dry cloth. (Care must be taken to prevent cleaning solvents from coming in contact with nonmetallic composites). When cleaned surfaces of the machine show signs of rusting before application of the final preservative, the cleaning process shall be confined to those specific small areas of the assembly. Alternatively, the solvent used for the final wiping operation shall contain a small amount (approximately 5 percent by volume) preservative oil conforming to P-10, type 1, grade 30. (It is extremely important that elapsed time between operations be held to an absolute minimum).

5.7.1.5 Disassembled components. Parts detached from the basic machine shall be cleaned and dried as specified in 5.7.1.2 and 5.7.1.3.

5.7.1.6 Steam and air lines. Drain all water accumulation and dry by injecting moisture-free air into the system.

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5.7.1.7 Maintenance or machine tools and tool accessories. All machine, maintenance, metal or woodworking, hand tools and any accessories there of shall be preserved and packaged in accordance with Level A requirements of PPP-P-40.

5.7.1.8 Electrical equipment. Electrical equipment, whether attached or detached, shall be cleaned, dried and preserved in accordance with Level A requirements of MIL-E-16298 or MIL-E-17555, which ever applies to the individual electrical item.

5.7.1.9 Electrical motors. Electric motors, whether attached or detached, shall be preserved and packaged in accordance with Level A requirements of MIL-E-16298, and applicable subparagraph for assembled machines.

5.7.1.10 Electrical control devices. Control units, whether attached or detached, shall be preserved and packaged in accordance with Level A requirements of MIL-E-17555.

5.7.1.11 Electrical wiring system. Exposed ends of wires, plugs openings, sockets, connector plugs, terminals and openings into switches and junction boxes shall be sealed with tape or cap-plugs conforming to PPP-T-60, type IV or MIL-C-5501, respectively. Electric cable shall be coiled to a minimum safe diameter and secured with cotton twine or tape to prevent uncoiling.

5.7.1.12 Hoses and fittings. Hoses and fittings which are not attached to the basic machine shall be cleaned, dried and preserved in accordance with Level A requirements of MIL-P-775.

5.7.1.13 Lubricating systems. The machine, with all lubricants. greases and other lubricating compounds either installed or included (see 5.8.2.9) shall have all vent openings in the lubrication systems temporarily capped, plugged or taped closed to prevent loss of lubricants during shipment and storage. Tags, warning that "ALL TEMPORARY CAPS, PLUGS OR TAPE MUST BE REMOVED BEFORE START-UP OR USE OF THE ITEM", shall be attached to all temporary caps, plugs or tape intended to be vented to the atmosphere during operation. In the event a lubrication reservoir cannot be covered and sealed, it shall appropriately be preserved and the lubricant shall be provided in a sealed container and shipped with the machine (see 5.8.2.9). A warning tag shall be placed in a conspicuous place near the filling area, warning the user to "INSTALL THE FURNISHED LUBRICANT BEFORE OPERATION OF THE EQUIPMENT".

5.7.1.14 Coolant systems. Coolant systems shall be drained and cleaned of all coolant residue and shall be preserved by flushing with P-3 preservative compound leaving a residual film of the preservative in the coolant lines and tank. Drain off the preservative and close the systems. Interior surfaces of the tank/reservoir which have not been covered or flooded with preservative while the machine is in operation shall be manually coated by brushing or spraying with similar type preservative.

5.7.1.15 High-speed spindle bearings. Reservoir shall be filled to normal operating level with manufacturer's lubricant and sealed closed for shipment and storage. If the system is of such design that the lubricant will leak out during shipment and storage, it shall be drained. The system shall then be preserved by filling with preservative oil P-10, type I, grade 30, drained, sealed and covered.

5.7.1.16 Hydraulic systems. Both open- and closed-type systems shall be filled to normal operating level with either the manufacturer's operative hydraulic oil or hydraulic oil conforming to MIL-H-6083. After filling, the machine shall be cycled to circulate the oil through the entire system. Prior to disconnecting the machine

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from the power source, place the piston(s), except opposing piston, in their innermost position. The oil shall remain in the reservoir, providing the reservoir can be sealed to prevent any leakage during shipment and storage. If it cannot be sealed to prevent leaking, it shall be drained and preserved and the hydraulic oil shall be provided in a sealed container and shipped with each machine.

5.7.1.16.1 Hydraulic systems for tracer mechanism. These systems shall be filled to a operating level with manufacturer's hydraulic oil or MIL-H-6083 and sealed or preserved as stated above in Hydraulic systems paragraph.

5.7.1.17 Grease fittings. Charge all grease fittings with manufacturer's compound or with preservative compound conforming to P-11.

5.7.1.18 Nonlubricated interior surfaces. Surfaces of internal gears which do not run in lubricant shall be thoroughly coated with P-7 preservative.

5.7.1.19 Air motors. Interior surfaces of air motors and air lines shall be coated with a preservative oil conforming to P-10, type I, grade 30 by injecting oil into the air stream while operating the machine until the oil appears at the exhaust ports. Air inlets and outlets shall be sealed with plastics caps, plugs or tape conforming to MIL-C-5501 or PPP-T-60, type IV, respectively. Air cylinders shall not be disassembled from the machine unless parts are vulnerable to damage during shipment and storage.

5.7.1.20 Gear cases. Fill the gear cases with normal lubricating oil. Shift gears into all possible positions while operating the machine. Make sure all surfaces are coated with lubricating oil and close and seal all openings to prevent any leakage during shipment and storage.

5.7.1.21 Sliding or friction surfaces. Surfaces of ways, gibs, driving gears, traversing and adjusting screws, and other critical, exterior operating machined surfaces shall be coated with P-2 preservative compound. Parts of the machine which are adjustable shall be locked into place by means provided and blocked to assure movement shall be kept to a minimum.

5.7.1.22 Non-critical surfaces. Exterior, non-critical, ferrous metal surfaces of the basic machine, which have not been painted or plated, shall be coated with type P-2 preservative and covered. Such places as legs, bolts, etc. from which the preservative need not be removed may be preserved with P-1 or P-19.

5.7.1.23 Bearings and journals. Grease lubricated bearings shall be preserved with type P-11 preservative. Oil lubricated journals shall be filled with normal oil lubricant or with lubricating oil P-10, type I, grade optional.

5.7.1.24 Drive belts and pulleys. Drive belts shall either be removed from the machine or released from tension. Belts removed from the machine shall be packaged in accordance with method IC-I or IC-3 criterion of MIL-P-116. Ferrous metal pulleys shall be coated with a rust-inhibiting primer conforming to TT-P-664.

5.7.1.25 Closed dial indicators. Closed dial indicators shall be covered with 1 inch thick uncompressed cushioning materials conforming to PPP-C-843, type II, class B. Secure cushioning material in place with tape conforming to PPP-T-60, type IV. Other type cushioning materials and tapes may be used provided the materials are water-resistant, non-corrosive, and will provide equivalent or better protection of those specified above.

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5.7.1.26 Drawers and door assemblies. Sliding surfaces of drawer guides shall be coated with P-11 preservative grease. Interior surfaces of door hinges shall be lubricated by the application of P-10, grade 30 preservative oil. In addition to mechanical locks or catches incorporated in the machine tool, secure doors and drawers with metal strapping or other devices which are suitable for the purpose intended. fiberboard pads shall be placed under metal strapping, to prevent marring the painted surfaces.

5.7.1.27 Closure of openings. Small openings (except vents and louvers installed for ventilation purposes) which will admit dust or water shall be sealed with tape conforming to PPP-T-60, type IV. Large openings shall be covered with waterproof barrier material conforming to class E-1 or L-4 of PPP-B-1055, or as a minimum, plastic material conforming to L-P-378, type and class optional. Secure materials in place with tape specified above. When very large openings are covered, or when location of openings render covering vulnerable to puncture, coverings shall be protected by applying wood, plywood or sheet metal over the materials. Open ends of all piping and fittings shall be plugged to prevent the entrance of foreign materials, using pipe fittings, plastic cap-plugs conforming to MIL-C-5501 or tape conforming to above.

5.7.1.28 Technical data. Technical data furnished with the machine shall be packaged in accordance with method IC-I criterion of MIL-P-116. The package shall be marked "TECHNICAL DATA" in bold black letters. The package shall be protected against puncture and abrasion. The data shall be either attached to the machine with tape conforming to PPP-T-60, type IV, placed in a storage compartment of the machine, or positioned in a consolidated box containing accessories or disassembled components.

5.7.1.29 Method II packaging. When specified (see 6.2), complex items which cannot be treated entirely externally with a contact preservative or where removal of the preservative would prevent timely use of the machine shall be packaged in accordance with Method II criterion of MIL-P-116. Accessories, maintenance tools, control units, and all internal surfaces of the machine shall be preserved and packaged as specified herein. Method II packs will bear precautionary labels as stated in MIL-STD-129. Upon receipt of a Method II package, it shall be inspected to assure the vacuum seal is intact before being placed in storage or shipped elsewhere.

5.7.1.30 Protection of preservatives. All exposed preserved surfaces of machines which are subject to being disturbed during the packing operation shall be loosely covered with barrier material conforming to MIL-B-121, type I, grade A. Not less than two layers of similar-type barrier material shall be used to separate wood or metal blocking and finished metal surfaces of the machine. When considered necessary, secure barrier in place with tape conforming to PPP-T-60, type IV.

5.7.1.31 Consolidated packaging. When size and weight permit, packaged accessories, maintenance tools, and small disassembled components shall be consolidated and placed in drawers and stowage compartments of the machine. When it is not feasible to stow parts as specified above, place parts in box(es) conforming to PPP-B-636, class WR or WWVR; PPP-B-640, class 2; PPP-B-601, domestic type; or PPP-B-621, class 1. Large bulky disassembled components shall be packed as specified in 5.7, as applicable.

5.7.1.32 Cushioning. Cushioning materials and methods of application shall conform to the criteria and guidelines of MIL-P-116 and MIL-STD-1186. In the event there are conflicting requirements, MIL-STD-1186 will have precedence.

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5.7.1.33 Movable parts. Operating heads, sliding tables, rams, counter-balances, and other movable parts shall be secured by mechanical devices incorporated in the machine. The devices alone do not always provide adequate support to prevent movement in the handling and transporting of the item and therefore must be augmented with appropriate wood blocking, metal tie rods used in conjunction with wood blocking, metal strapping, or other devices which will immobilize the part. Handwheels and levers, and similar controls shall be secured in position with common annealed, wire or steel strapping.

5.7.1.34 Positioning assemblies. Assemblies having vertical movement shall be blocked and braced in a lowered position; those capable of moving traversely shall be centered for blocking and bracing.

5.7.1.34.1 Tables. Tables and other components moving on ball bearings or other types of high efficiency, low-friction ball or roller bearing assemblies shall be removed, or blocked, and all components treated in such a manner that neither the way(s), surface(s), nor the anti-friction devices will be subject to brinelling or other damage.

5.7.1.34.2 Counterweights. Counterweights shall be locked and blocked in a manner which will release all tension from supporting cables, chains and springs. Concealed counterweights suspended in closed columns shall be removed or blocked to prevent vertical or horizontal movement.

5.7.2 Level C preservation. Preservation of metal and wood-working machinery shall conform to the requirements of MIL-STD-1190.

5.8 Packing. Packing shall be Level A, B or C, as specified in the contract.

5.8.1 General requirements for Levels A and B. When weight, size and design permit, machines shall be assembled when being prepared for shipment. When it is not considered feasible to ship a machine assembled, components shall be packed in accordance with weight limitations specified herein.

5.8.2 Level A.

5.8.2.1 Machinery weighing less than 2,500 pounds. Unless otherwise specified (see 6.2), machinery including accessories, weighing less than 2,500 pounds, shall be packed in cleated plywood boxes conforming to PPP-B-601, overseas type; nailed wood boxes conforming to PPP-B-621, class 2, style 2, 3, 4 or 7; skidded boxes conforming to MIL-B-26195, type II, style A, class optional, with plywood superstructure. Selection of containers shall be governed by the size and weight limitations of the applicable container specification. Machines designed with mounting holes in areas that can withstand rough handling without breakage shall be bolted to an auxiliary base before being packed in a box conforming to PPP-B-601 or PPP-B-621. Contents of boxes shall be arranged, anchored, blocked, braced and cushioned in accordance with MIL-P-116 and MIL-STD-1186. In the event there are conflicting requirements, MIL-STD-1186 will have precedence. Top-heavy machines shall be packed as specified in 5.8.2.5. Box closure and strapping requirements shall be in accordance with the appendix to the applicable box specification.

5.8.2.2 Machinery weighing 2,500 to 30,000 pounds. Unless otherwise specified (see 6.2), machines weighing 2,500 to 30,000 pounds shall be packed individually in crates conforming to MIL-C-104, type I, class 2, style A. To conserve cubage, safeguard components of the machine, and to comply with weight and dimensional

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limitations of the crate specification, subassemblies removed through disassembly and control units may be packed in separate shipping containers. When packed separately, items weighing less than 2,500 pounds shall be packed as specified in 5.8.2.1. Arrangement of contents, anchoring, blocking, bracing and cushioning shall be in accordance with the appendix to the crate specification, MIL-STD-1186, and as specified herein. MIL-STD-1186 will have precedence when there are conflicting requirements. When practicable, to facilitate loading into common carriers, the height of the crated machine shall be limited to 9 feet 8 inches.

5.8.2.3 Buttress blocking. Hold down bolts on machines weighing over 8,000 pounds shall be reinforced with backup buttress blocking extending from the machine base to the skid header; blocking shall be parallel to skid runners and in line with bolt holes on the machine. The blocks shall be similar to that used for the skid header. Carriage or step bolts shall be utilized to secure the block to the skid base.

5.8.2.4 Unsupported columns. Unsupported columns of the machine shall be reinforced by placing snug-fitting blocks between the columns and the skid platform or machine base, as applicable. Blocks extending from the column to the skid base shall be fitted so that tightening of the hold down bolts will not place undue stress on the column(s).

5.8.2.5 Top-heavy machinery. To safeguard equipment and personnel, top-heavy machine tools may be packed in a horizontal position, design permitting. Blocking and bracing shall be located at points of the machine with sufficient rigidity to carry the machine weight. Crates for top-heavy items requiring upright skidding due to design, structural weakness, etc., shall be sized to provide stability in handling and transit. The length and width of the skid platform shall extend beyond the machine a distance equivalent to the height of the center of gravity of the machine above the base.

5.8.2.6 Leg-type machines. Leg-type machines with slotting holes for mounting shall be reinforced by placing a nominal 2- by 4-inch wood block against the leg opposite the slot opening and nailing to the skid base.

5.8.2.7 Bolting. Machines shall be secured to the skid base by not less than four bolts. When sufficient bolting points are not available, J-clamps or tie rod assemblies as specified in 5.7.1.33, or other devices, shall be used to provide the minimum of four holding points. Bolts shall pass completely through the skid runner or sleeper, as applicable, and the base flooring.

5.8.2.8 Consolidated containers. Consolidated containers specified in 5.7.1.31 shall be placed inside the container housing the machine. The boxes shall be restrained by the use of metal strapping of sufficient strength or tie rod assemblies.

5.8.2.9 Operational oils. Operational oils furnished with the machine, but not placed in the machine stowage receptacles, shall be stowed in metal drum(s) conforming to PPP-D-705, type I or II or PPP-D-729, type I or II. Drums shall be positioned upright inside the shipping container housing the machine. Sufficient blocking shall be applied to immobilize the drum(s) inside the shipping container. Alternatively, when the drum(s) tend to increase the cubage of the shipping container, the drum(s) may be shipped as individual containers.

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5.8.2.10 Machines exceeding crate limitations. Unless otherwise specified (see 6.2), machines exceeding the weight and size limitations of MIL-C-104 shall be packed in a similarly designed crate. The crate's structural members and anchoring devices shall be increased in number and size as needed to accommodate the load to be packed therein.

5.8.2.11 Level B. Machines weighing 16,000 pounds or less shall be packed for shipment utilizing boxes conforming to PPP-B-601, domestic type, style A, B, I, or J; PPP-B-621, class 1, style 2, 3, 4, or 7; MIL-B-26195, type I, style A; or crates conforming to MIL-C-52950 or MIL-C-3774. Selection of the container shall be governed by the size and weight limitations of the applicable container specification. Machines designed with mounting holes in areas that can withstand rough handling without breakage shall be bolted to an auxiliary base before being packed in a box conforming to PPP-B-601 or PPP-B-621. Arrangement of contents, anchoring, cushioning, blocking, bracing and container closure shall be the same as for level A (see 5.8.2). Items packed in open crates shall be covered with a waterproof shroud conforming to MIL-STD-1186. Machines exceeding 16,000 pounds shall be packed as specified in 5.8.2.2.

5.8.2.12 Level C packing. Packing of metal and woodworking machinery shall conform to the requirements of MIL-STD-1190. When authorized by the transportation media and when safe handling with mechanical equipment is provided, the machines may be mounted on skids or a skid base in lieu of being boxed or crated. Openings in skids shall be a minimum of 3 inches in height to provide access for forklift tines. Machines not packed in fully sheathed containers shall be completely covered with a waterproof plastic shroud. The shroud shall be secured to the base of the machine. Polyvinyl chloride (PVC) and ethylene vinyl acetate (EVA) shall not be acceptable as an intimate wrap or shroud material due to the possible corrosive effect of PVC or EVA vapors on the shrouded item and its component parts.

5.9 Marking. All marking shall be in accordance with MIL-STD-129. Bar code markings are required and shall be in accordance with MIL-STD-129. Center of balance markings and arrow markings, indicating THIS SIDE UP, shall be applied to each shipping container for all levels for the protection of the machinery and personnel.

5.10 Rough handling. The completed unit pack, shipping container(s) and all contents shall be capable of withstanding the shipping and handling conditions expected from fork lift handling, truck transport or superimposed load. The container, its contents and the packaging shall not be damaged, functionally or physically, or items displaced by the appropriate rough handling test(s).

5.11 Workmanship. Workmanship shall be such that when the proper procedure is followed, materials and equipment being processed will be provided the maximum protection against corrosion and deterioration consistent with the shipment hazards known or anticipated, and suitable for the term of storage expected.

6 NOTES

(This section contains information of a general explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The blast cleaning machine covered by this specification is intended for use in removing rust, scale, and burrs from metal shapes, for the preparation of metal surfaces for plating, and for cleaning castings and forgings.

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6.2 Acquisition requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.
d. First article when required (see 3.1).
e. Special safety devices when required (see 3.4.6 and 6.5).
f. Type of endless belt conveyor required (see 3.6.3).
g. Dust collector if different (see 3.6.6).
h. Wiring if different (see 3.6.7).
i. Motors if different (see 3.6.7.1).
j. Spare parts when required (see 3.7).
k. Identification marking if different (see 3.9).
l. First article inspection if different (see 4.3).
m. Quality conformance inspection if different (see 4.4).
n. Packaging first article, when required (see 5.1).
o. Level of preservation required (see 5.7).
p. Method II packaging when required (see 5.7.1.29).
q. Packing of machinery weighing less than 2,500 pounds if different (see 5.8.2.1).
r. Packing of machinery weighing 2,500 to 30,000 pounds if different (see 5.8.2.2).
s. Packing of machines exceeding the weight and size limitations of MIL-C-104 if different (see 5.8.2.10)

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instruction for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.4 First article. When a first article inspection is required, the blast cleaning machine should be a first article sample, a first production item, a sample selected from the first run of production items or it may be a standard production item from the contractor's current inventory as specified in 4.3. The first article should consist of a specific number of units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering products which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to reply on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and controls of noise levels, radiation, electromagnetic emissions, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.6 Repair part manuals. Repair part identification, in repair party manuals, should include the prime contractor's name, nomenclature, part number, and price. Repair part identification should also include the name of the original

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equipment/part manufacturers, Contractor and Government Entity (CAGE), nomenclature, and part number immediately after the prime contractor's part identification. High mortality rate items shall be so identified as shown in MIL-M-7298.

6.7 Subject term (key word) listing.

Abrasive cleaning
Blast cleaning machine
Sand blasting machine
Tumble cleaning machine

6.8 Blast cleaning machine changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue.

Custodian:
Army - AL

Preparing activity:
Army - AL

Review activity:
Army - AL
DLA - CS

Project No. 4940-A608

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-B-45081E(AL) Blast Cleaning Machine: Self-Contained, Tumble Type, with Dust Collector

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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